EXPERIMENT 1: STUDY OF DIODE CHARACTERISTIC

INTRODUCTION

The semiconductor diode is created by simply joining an n-type and a p-type material together nothing more just the joining of one material with a majority carrier of electrons to one with a majority carrier of holes.

The P-N junction supports uni-directional current flow. If +ve terminal of the input supply is connected to anode (P-side) and –ve terminal of the input supply is connected to cathode (N- side), then diode is said to be forward biased.

In this condition the height of the potential barrier at the junction is lowered by an amount equal to given forward biasing voltage. Both the holes from p-side and electrons from n-side cross the junction simultaneously and constitute a forward current (**injected minority current** – due to holes crossing the junction and entering N-side of the diode, due to electrons crossing the junction and entering P-side of the diode).

Assuming current flowing through the diode to be very large, the diode can be approximated as short-circuited switch. If –ve terminal of the input supply is connected to anode (p-side) and +ve terminal of the input supply is connected to cathode (n-side) then the diode is said to be reverse biased.

OBJECTIVE

1. To plot V-I characteristic of 1N4007 diode.
2. To find the knee voltage of 1N4007 diode.
3. To find the reverse breakdown voltage of 1N4007 diode.
4. To plot V-I characteristic of 1N4007 diode using DC sweep.

PART A: P-N JUNCTION DIODE CHARACTERISTIC

EQUIPMENT AND COMPONENT

1. Variable DC Power Supply
2. Resistor (5Ω)
3. Diode (IN4007)
4. Multimeter
5. Connecting wires

PROCEDURE

Circuit diagram

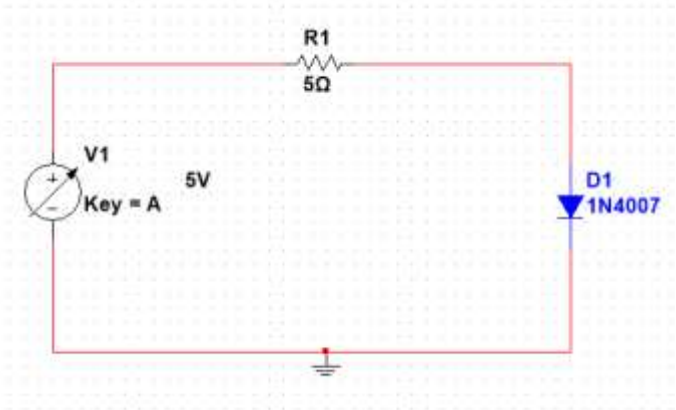


Figure 1: Forward biased condition

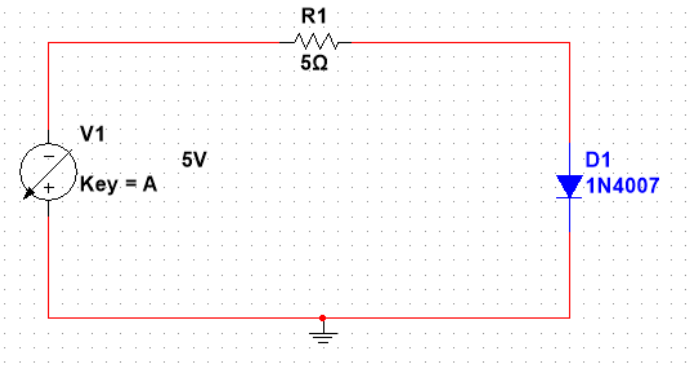


Figure 2: Reversed biased condition

Forward Biased Condition:

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM103_Experiment 1_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 12 to construct your forward biased circuit as shown in figure 1.
4. Click Place > Component > Group > Sources > SIGNAL_VOLTAGE_SOURCES > DC_INTERACTIVE_VOLTAGE > OK. Place it on the screen.
5. Double click DC_INTERACTIVE_VOLTAGE, set Max. value 8V, Min. value 0V and increment step 0.05V (0.625%).
6. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.
7. Set your RESISTOR value to 5Ω. Double click on RESISTOR > Value > Resistance > 5 Ω > OK.
8. To rotate the RESISTOR > Click RESISTOR > Ctrl + R **or** Right click > Rotate 90^0 Clockwise/Counterclockwise.
9. Click Place > Component > Group > Diodes > DIODE >Type 1N4007 > OK. Place it on the screen.
10. Connect all component as shown in Figure 1.
11. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in series connection with the circuit for current,I_d measurement as in figure 3.
12. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in parallel connection with the 1N4007 diode for voltage, V_d measurement as in figure 3.

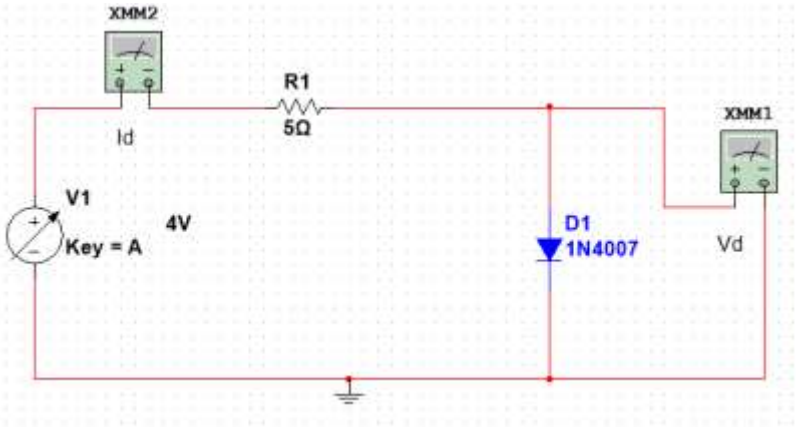


Figure 3: Forward biased condition with Multimeter

13. Run the simulation.

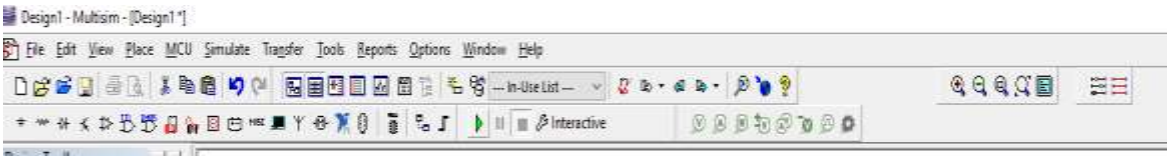


Figure 4: Run button

14. Record the corresponding reading of Vd (V) and Id (A) in Table 1.
15. Tabulate different forward current obtained for different forward voltages.
16. Plot the V-I characteristic graph for forward biased condition.

Reversed Biased Condition:

1. Follow Step 2 until Step 11 to construct your reverse biased circuit as shown in figure 2.
2. Click Place > Component > Group > Sources > SIGNAL_VOLTAGE_SOURCES > DC_INTERACTIVE_VOLTAGE > OK. Place it on the screen.
3. Double click DC_INTERACTIVE_VOLTAGE, set Max. value 1500V, Min. value 0V and increment step 37.5V (2.5%). Ensure the polarity in reversed.
4. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.
5. Set your RESISTOR value to 5Ω. Double click on RESISTOR > Value > Resistance > 5 Ω > OK.
6. To rotate the RESISTOR > Click RESISTOR > Ctrl + R **or** Right click > Rotate 90^0 Clockwise/Counterclockwise.
7. Click Place > Component > Group > Diodes > DIODE >Type 1N4007 > OK. Place it on the screen.
8. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in series connection with the circuit for current,I_d measurement as in figure 5.
9. Click Simulate > Instruments > Multimeter. Place it on the screen. Connect Multimeter in parallel connection with the 1N4007 diode for voltage, V_d measurement as in figure 5.
10. Connect the circuit as shown in figure 2.
11. Record the corresponding reading of Vd (V) and Id (A) in Table 2.
12. Tabulate different reverse currents obtained for different reverse voltages.
13. Plot the V-I characteristic graph for reverse biased condition.

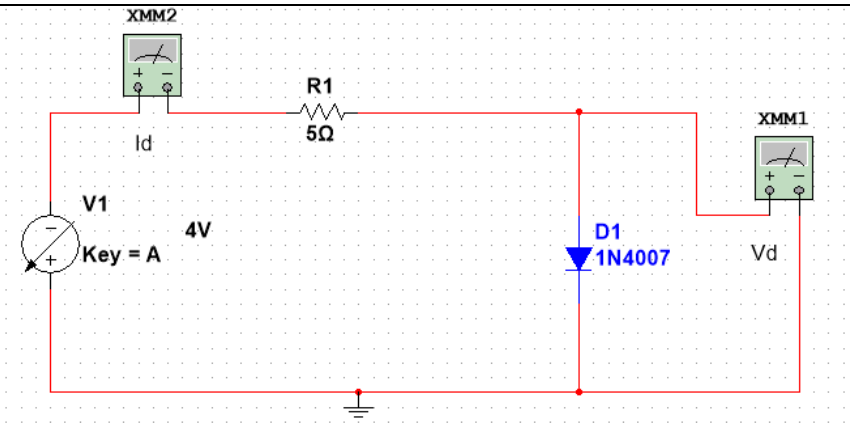


Figure 5: Reversed biased condition with Multimeter

RESULT

Table 1: 1N4007 diode in forward biased conditions

Voltage Supply, Vs	Voltage across diode, Vd (V)	Current flow through diode, Id (A)
0		
0.05		
0.1		
0.2		
0.3		
0.4		
0.5		
0.6		
0.7		
0.8		
0.9		
1		
1.5		
2		
2.5		
3		
3.5		
4		
4.5		
5		

Table 2: 1N4007 diode in reversed biased conditions

Voltage Supply, Vs	Voltage across diode, Vd (V)	Current flow through diode, Id (A)
0		
37.5		
75		
150		
225		
300		
375		
450		
525		
600		
675		
750		
825		
900		
975		
1.05k		
1.13k		
1.20k		
1.28k		
1.35k		

PART B: P-N JUNCTION DIODE CHARACTERISTIC BY USING DC SWEEP

EQUIPMENT AND COMPONENT

1. DC Power Supply
2. Resistor (5Ω)
3. Diode (IN4007)
4. Connecting wires

PROCEDURE

Forward Biased Condition:

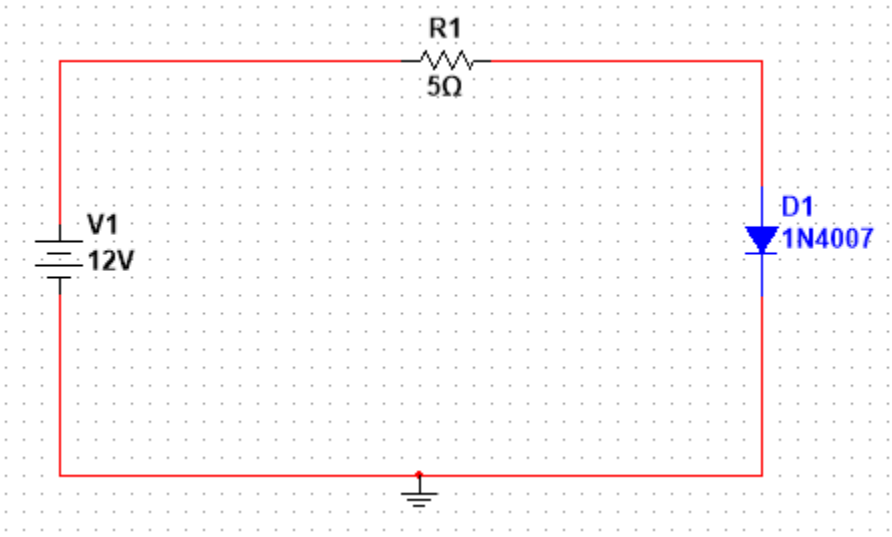


Figure 6: Forward biased condition

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM103_Experiment 1_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 10 to construct your forward biased circuit as shown in figure 6.
4. Click Place > Component > Group > Sources > POWER_SOURCES > DC_POWER > OK. Place it on the screen.
5. Double click DC_POWER, set 12V.
6. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.

7. Set your RESISTOR value to 5Ω. Double click on RESISTOR > Value > Resistance > 5 Ω > OK.
8. To rotate the RESISTOR > Click RESISTOR > Ctrl + R **or** Right click > Rotate 90^0 Clockwise/Counterclockwise.
9. Click Place > Component > Group > Diodes > DIODE >Type 1N4007 > OK. Place it on the screen.
10. Connect the circuit connection as in figure 6.
11. Run the simulation. Click > SIMULATE at toolbar menu > Analyses and Simulation > DC Sweep.
12. DC Sweep setting. Analysis parameter > Sources 1 > Start Value = 0V > Stop Value = 0.8V > Increment value = 0.01V.
13. DC Sweep setting. Output > I (D1[D1]) > Save > Run.
14. Record the graph.

Reversed Biased Condition:

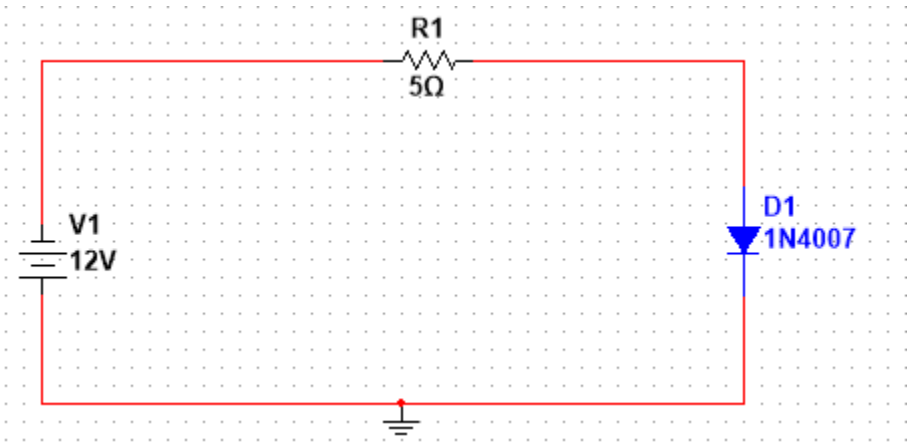


Figure 7: Reversed biased condition

1. Run Multisim 14.0 Software.
2. Click File > Save As > COM103_Experiment 1_YOUR ID_YOURNAME > Save.
3. Follow Step 4 until Step 9 to construct your forward biased circuit as shown in figure 7.
4. Click Place > Component > Group > Sources > POWER_SOURCES > DC_POWER > OK. Place it on the screen.
5. Double click DC_POWER, set 12V. Reversed the polarity of the DC power source.
6. Click Place > Component > Group > Basic > RESISTOR > OK. Place it on the screen.

7. Set your RESISTOR value to 5Ω. Double click on RESISTOR > Value > Resistance > 5 Ω > OK.
8. To rotate the RESISTOR > Click RESISTOR > Ctrl + R **or** Right click > Rotate 90^0 Clockwise/Counterclockwise.
9. Click Place > Component > Group > Diodes > DIODE >Type 1N4007 > OK. Place it on the screen.
10. Connect the circuit connection as in figure 7.
11. Run the simulation. Click > SIMULATE at toolbar menu > Analyses and Simulation > DC Sweep.
12. DC Sweep setting. Analysis parameter > Sources 1 > Start Value = -1V > Stop Value = 1V > Increment value = 0.01V.
13. DC Sweep setting. Output > edit expressions > - I (D1[D1]) > Save > Run.
14. Record the graph.

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. From the graph in part A, what is the value for Knee voltage of 1N4007 diode?
3. From the graph in part A, what is the value for reverse breakdown voltage of 1N4007 diode?
4. What is the difference between ideal and practical diode characteristic?
5. What is the knee voltage? Give potential barrier values for Ge and Si.

Components	Poor (1)	Moderate (2)	Good (3)	Excellent (4)	Score
Abstract/Summary (5 Marks)	Several major aspects of the experiment are missing; student displays a lack of understanding about how to write an abstract	Abstract misses one or more major aspects of carrying out the experiment or the results	Abstract references most of the major aspects of the experiment, some minor details are missing	Abstract contains reference to all major aspects of carrying out the experiment and the results, well written	
Introduction (5 Marks)	Very little background information provided or information is incorrect	Some introductory information, but still missing some major points	Introduction is nearly complete, missing some minor points	Introduction is complete and well written; provides all necessary background principles for the experiment	
Objective (5 Marks)	The objectives of the experiment are erroneous or irrelevant.	The objectives of the experiment are partially identified or vague.	The objectives of the experiment are mostly identified.	The objectives of the experiment are clearly identified and well stated.	
Experimental Procedure (10 Marks)	Missing several important experimental details	Missing some important experimental details	Important experimental details are covered, some minor details missing	All experimental details are covered	
Results: data, figures, graphs, tables, etc. (30 Marks)	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions or numbers, units missing or incorrect, etc. Important calculation steps (if any) are missing.	Most figures, graphs, tables OK, some still missing some important or required features. Some important calculation steps (if any) are missing.	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved. All-important calculation steps (if any) are included but minors error is present.	All figures, graphs, tables are correctly drawn, are numbered and contain titles/captions. All important calculation steps (if any) have been included and accurately presented.	
Discussion (30 Marks)	Very incomplete or incorrect interpretation of trends and comparison of data indicating a lack of understanding of results.	Some of the results have been correctly interpreted and discussed; partial but incomplete understanding of results is still evident	Almost all of the results have been correctly interpreted and discussed, only minor improvements are needed	All-important trends and data comparisons have been interpreted correctly and discussed, good understanding of results is conveyed	
Conclusions (5 Marks)	Conclusions missing or missing the important points	Conclusions regarding major points are drawn, but many are misstated, indicating a lack of understanding	All-important conclusions have been drawn, could be better stated	All-important conclusions have been clearly made, student shows good understanding	
References (5 Marks)	References are listed in wrong format. References are not cited in the text. No reference cited	Some references are cited in the text and listed in a references cited section, however, formatting is incorrect for all references, or insufficient or inappropriate eeferences are used.	Sufficient and appropriate sources cited. Sources are cited in the text. Most references are formatted appropriately and consistently.	Sufficient and appropriate sources cited. Sources are cited appropriately in the text (Author Date). All references are formatted appropriately and consistently	
Appearance and Formatting (5 Marks)	Sections out of order, too much handwritten copy, sloppy formatting	Sections in order, contains the minimum allowable amount of handwritten copy, formatting is rough but readable	All sections in order, formatting generally good but could still be improved	All sections in order, well formatted, very readable	
TOTAL SCORE					**/100**